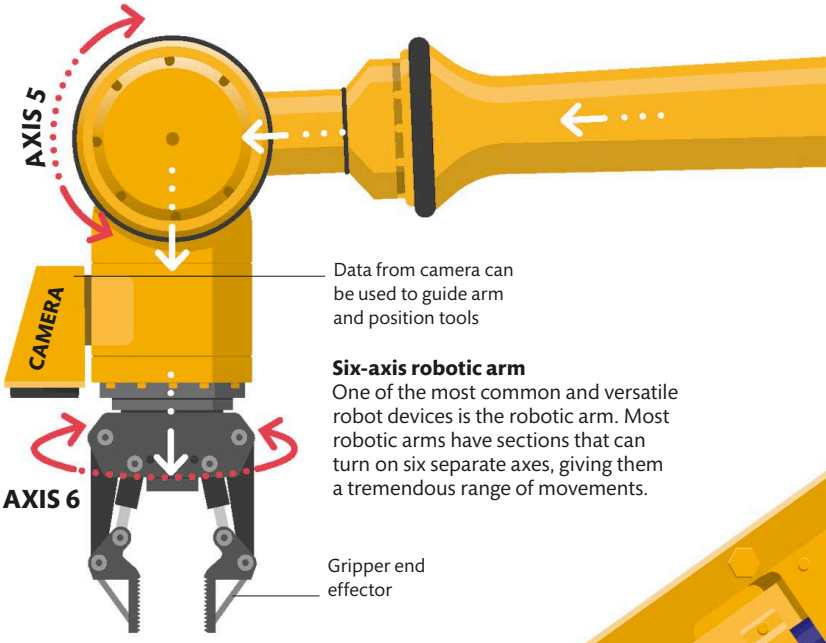




End effectors
Many different kinds of tools can be fitted to the hand of a robotic arm, more correctly called the end effector. The most common is a gripper that can pick up, move, and drop small objects.



Six-axis robotic arm
One of the most common and versatile robot devices is the robotic arm. Most robotic arms have sections that can turn on six separate axes, giving them a tremendous range of movements.

How do robots work?

A robot is a computer-controlled machine that can do a range of tasks with little or no human intervention. Robots are used in factories and warehouses, in education, by the military, in the home, and just for fun.

How robots move

The parts of a robot that enable it to move and to manipulate objects are called actuators. The computer that controls a robot sends precise electric currents that make the actuators work. Most actuators are driven by a kind of electric motor called a stepper motor (see opposite). This kind of motor turns in small steps—making it possible for the parts of a robot to move precisely into the desired position. Some robots can also move around, using wheels, caterpillar tracks, or even legs.

CAN A ROBOT BE HACKED?

Yes, hackers can rewrite the programs in the computer that controls a robot. Making sure robots are safe and secure will become an important issue as robots become more common.

